

BANK LOAN ANALYTICS REPORT

SQL + Power BI Portfolio Project

Prepared by Sujal Mondal

Loan portfolio performance, risk patterns and business recommendations

Project focus

I built this project to move beyond a basic dashboard and use SQL to answer portfolio questions, then use Power BI to present the results in an interactive way. The report below summarizes the analysis, the main patterns I found in the data, and practical areas a lending team could investigate further.

Source files used: financial_loan.csv, SQL EDA queries, SQL business-question queries, Power BI dashboard and the supplied problem statement.

1. Project Overview

A practical portfolio analysis of loan applications, funding, repayment and risk.

Area	What I worked on
Dataset	38,576 loan records with 23 fields covering loan, borrower, status and payment information.
SQL analysis	EDA checks plus business questions covering volume, repayment, grade, state, purpose, term and verification.
Power BI	Two-page interactive dashboard: Summary + Overview, with KPI cards, trend analysis, risk breakdowns, etc.
Main objective	Understand portfolio growth, repayment performance and where credit risk is concentrated.

The original problem statement is centered on analyzing loan portfolio performance, tracking good vs bad loans and repayment behavior, monitoring funded amount, repayments, interest rate and DTI, and building an interactive Power BI report. [filecite\turn0file2\L30-L39](#)

Data model used in SQL

The SQL table contains fields such as issue date, loan ID, purpose, verification status, grade, home ownership, state, loan status, employment information, annual income, DTI, interest rate, loan amount and total payment. [filecite\turn0file0\L3-L24](#)

Analysis flow

Raw CSV → SQL data checks → SQL EDA → business questions → Power BI model/dashboard → findings → recommendations

2. Data Quality & SQL Work

The first part of the project was about making sure the dataset was understandable before interpreting the dashboard.

Check	Result	Why it matters
Records	38,576	Defines the portfolio size used throughout the report.
Columns	23	Covers loan, borrower, status and payment attributes.
Duplicate Loan IDs	0	No repeated loan IDs were found.
Duplicate Member IDs	0	No repeated member IDs were found in the dataset.
Completely duplicated rows	0	No identical rows were found.
Missing values	EMP_TITLE: 1,438; other checked fields 0	Employment title is the only field with missing values in the CSV.
Date range	01-Jan-2021 to 12-Dec-2021	The portfolio covers the 2021 period present in the file.

The SQL EDA includes checks for record count, columns, duplicate IDs, missing values, date range, loan status, purpose, grade, sub-grade, home ownership, verification, term, state and basic statistics. It also extends into monthly trends, grade-level analysis, charged-off loans and good/bad categorization.

Data quality note: EMP_TITLE has 1,438 missing values. I did not treat that as a reason to remove records because employment title is not required for the core portfolio KPIs or the risk cuts used in this report.

Business questions covered

The second SQL script asks 15 business questions, including total applications and loan amount, repayments, charged-off grades and states, interest rate by grade, monthly portfolio movement, DTI by grade, bad-loan rate by grade/purpose/term/verification and the Good/Bad/Current mix.

3. Portfolio Snapshot

The headline numbers from the dataset and the Power BI dashboard.

38,576 Applications	435.8M Funded amount	473.1M Total amount received	12.05% Average interest rate	13.33% Average DTI
-------------------------------	--------------------------------	---	---	------------------------------

Good / Bad view

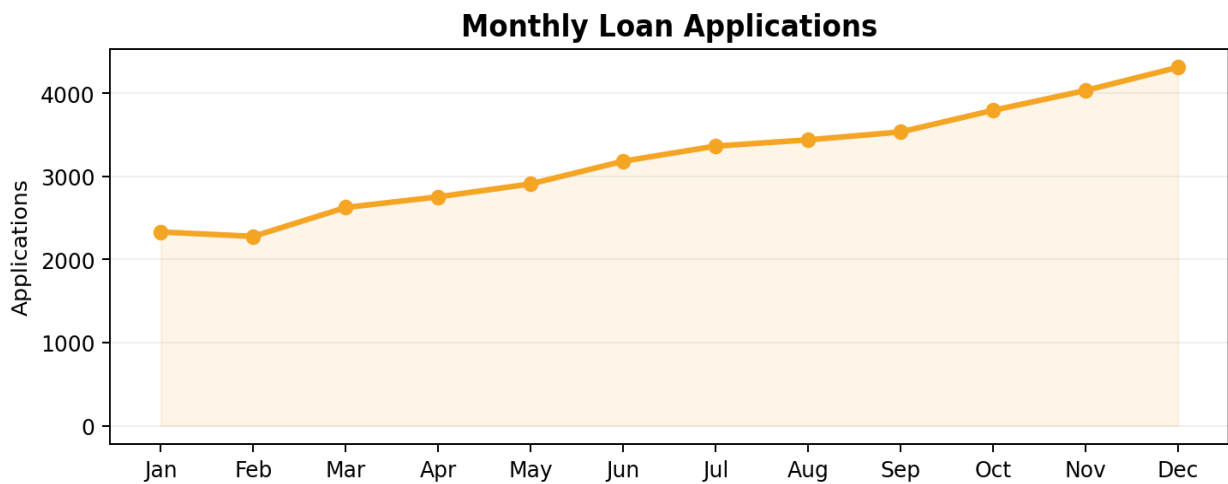
Dashboard view	Applications	Share	Loan amount
Good (Fully Paid + Current)	33,243	86.18%	370.2M
Bad (Charged Off)	5,333	13.82%	65.5M
Current inside dashboard Good	1,098	2.85%	Included in Good card

Important definition note: the dashboard's 86.18% Good figure is consistent with treating every non-charged-off loan as Good (33,243 = Fully Paid + Current). The SQL business-question query keeps Fully Paid, Charged Off and Current as three separate categories. I have kept that distinction visible rather than hiding it.

The dataset contains 32,145 fully paid loans, 5,333 charged-off loans and 1,098 current loans. Total payments are 37.3M higher than original funded loan amount (8.56% of funded amount). This difference should not be read as profit because the dataset does not isolate interest income, fees, operating costs or timing effects.

4. Portfolio Growth & Trends

Application volume increased consistently through the year.



Applications increased from 2,332 in January to 4,314 in December, an increase of 85.0%. Monthly funded loan amount increased from 25.0M to 54.0M, up 115.7%. The Power BI Overview page makes this pattern easy to see with the monthly application line chart.

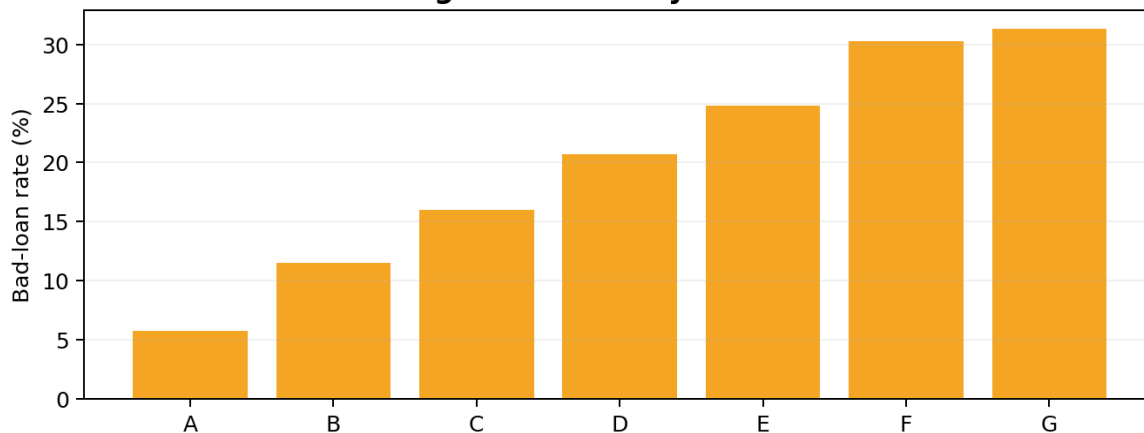
Monthly application table

Month	Applications	Loan amount	Amount received
Jan	2,332	25.0M	27.6M
Feb	2,279	24.6M	27.7M
Mar	2,627	28.9M	32.3M
Apr	2,755	29.8M	32.5M
May	2,911	31.7M	33.8M
Jun	3,184	34.2M	36.2M
Jul	3,366	35.8M	38.8M
Aug	3,441	38.1M	42.7M
Sep	3,536	40.9M	44.0M
Oct	3,796	44.9M	49.4M
Nov	4,035	47.8M	50.1M
Dec	4,314	54.0M	58.1M

5. Credit Risk Findings

The strongest pattern in the data is the increase in charged-off rate as risk grade worsens.

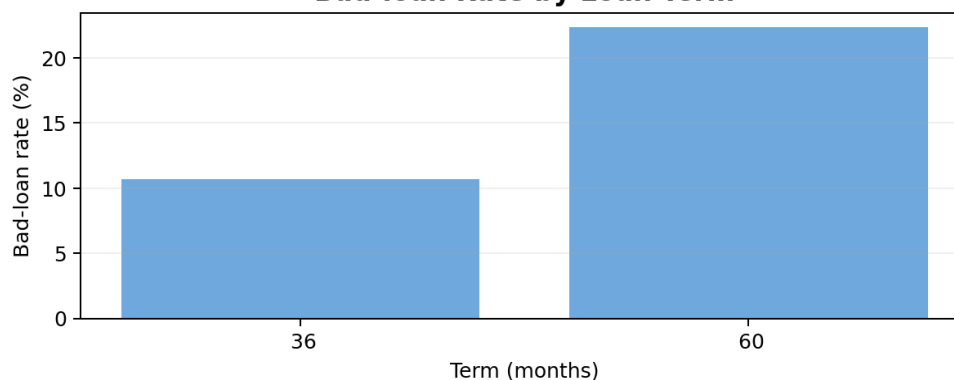
Charged-Off Rate by Loan Grade



Grade	Applications	Bad rate	Avg. interest
A	9,689	5.70%	7.35%
B	11,674	11.50%	11.03%
C	7,904	16.02%	13.55%
D	5,182	20.69%	15.71%
E	2,786	24.80%	17.71%
F	1,028	30.25%	19.74%
G	313	31.31%	21.40%

The observed bad-loan rate rises from 5.70% in Grade A to 31.31% in Grade G. Average interest rate also rises from 7.35% in Grade A to 21.40% in Grade G. This makes grade a useful segmentation field for portfolio monitoring.

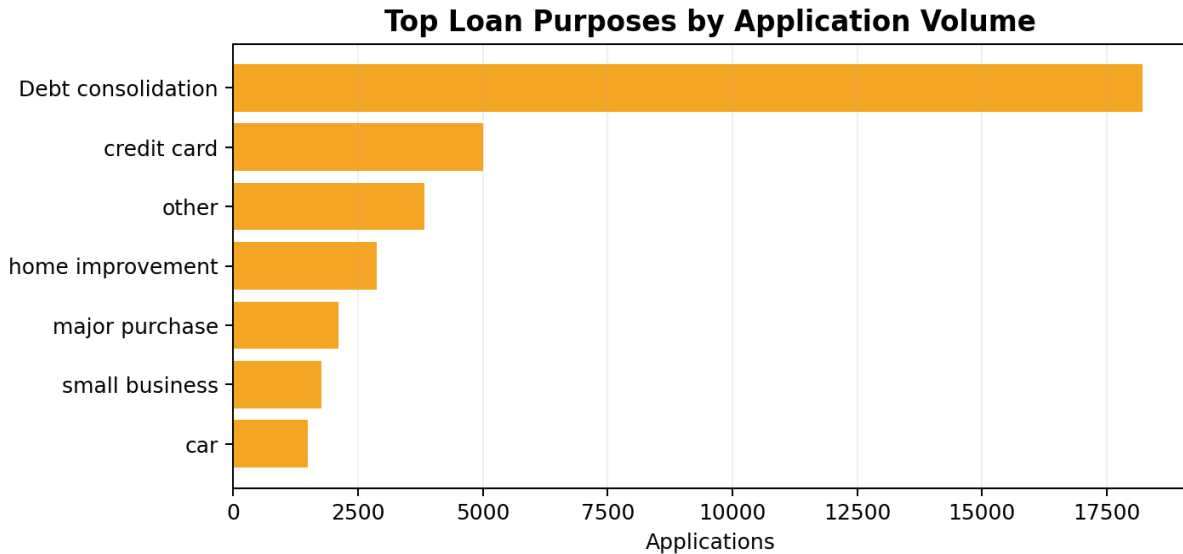
Bad-loan Rate by Loan Term



Loan term also shows a clear difference: 36-month loans have a 10.71% charged-off rate, while 60-month loans are at 22.34%. The dashboard shows that 36 months accounts for about 73.2% of applications and 60 months about 26.8%.

6. Purpose, State & Verification Findings

Volume and risk are not concentrated in exactly the same places.



Debt consolidation is by far the largest purpose with 18,214 applications. It also contributes the largest charged-off amount at about 36.26M. Small business has a much smaller volume (1,776 applications) but the highest observed bad-loan rate among the listed purposes at 25.62%.

Purpose	Applications	Bad rate	Charged-off amount
Debt Consolidation	18,214	14.55%	36.3M
Credit Card	4,998	10.16%	6.7M
Other	3,824	15.35%	5.0M
Home Improvement	2,876	11.37%	3.9M
Major Purchase	2,110	9.76%	1.9M
Small Business	1,776	25.62%	6.7M
Car	1,497	10.35%	1.1M
Wedding	928	9.27%	1.0M

Highest portfolio exposure by state

State	Applications	Funded amount	Bad rate	Bad amount
CA	6,894	78.5M	15.33%	13.1M
NY	3,701	42.1M	12.65%	5.8M
TX	2,664	31.2M	11.30%	3.8M
FL	2,773	30.0M	17.27%	5.6M
NJ	1,822	21.7M	14.98%	3.5M
IL	1,486	17.1M	12.92%	2.4M

California has the highest funded exposure at about 78.48M and the highest charged-off amount at about 13.12M. Florida has a 17.27% bad-loan rate in this dataset, higher than California's 15.33%. These are portfolio monitoring observations, not causal conclusions.

Verification segmentation

Verification status	Applications	Bad loans	Bad rate
Not Verified	16,464	2,015	12.24%
Source Verified	9,777	1,382	14.14%
Verified	12,335	1,936	15.70%

Verified loans show a 15.70% observed bad-loan rate versus 12.24% for Not Verified. This should be investigated as a mix/selection pattern rather than interpreted as verification causing higher risk.

7. Business Findings & Recommendations

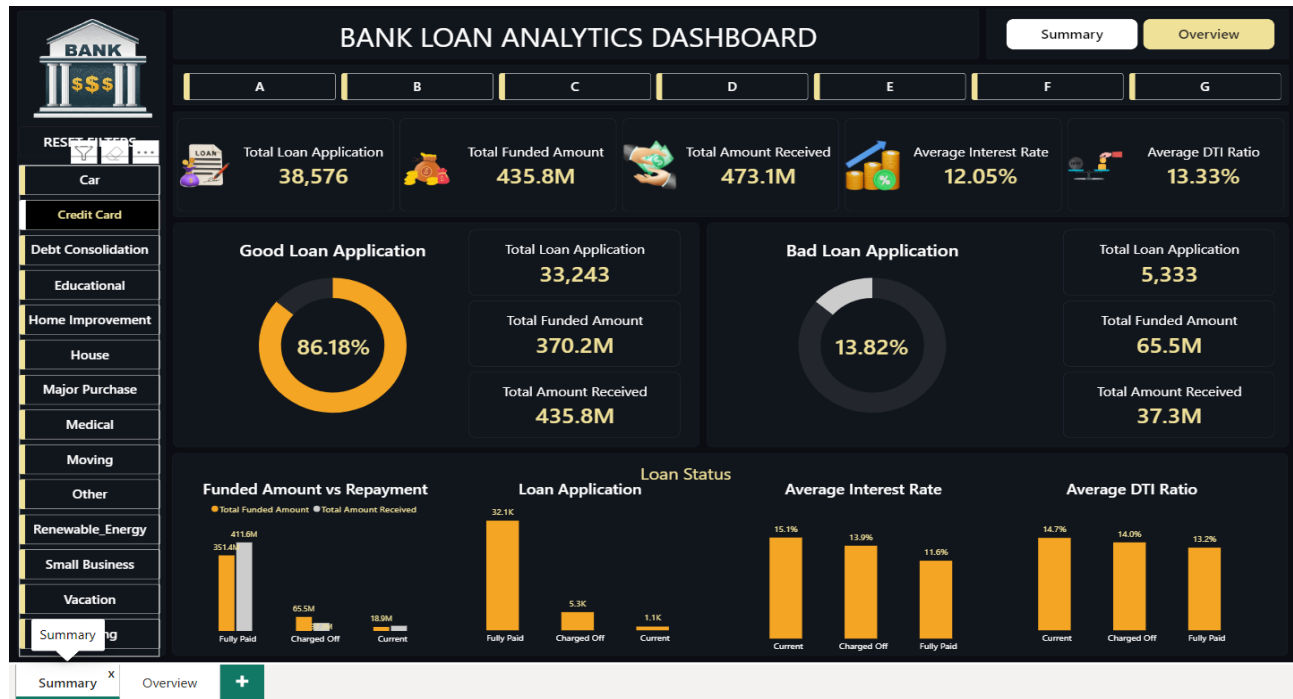
These recommendations come directly from the patterns found in the dataset.

01	Use grade-based risk monitoring Grades E-G have materially higher charged-off rates than A-C. Add a monthly risk view by grade, with special review thresholds for the higher-risk grades.
02	Review 60-month lending separately The 60-month segment has a 22.34% bad-loan rate versus 10.71% for 36 months. Compare approval mix, borrower profile and repayment behavior before changing policy.
03	Create an early-warning view for debt consolidation Debt consolidation has the largest application volume and the largest charged-off amount. Monitor this segment using grade, DTI, term and state together rather than volume alone.
04	Investigate small-business loans Small business has a 25.62% bad-loan rate. Because the segment is smaller, review individual risk drivers and sample size before making broad policy changes.
05	Focus geographic monitoring on exposure + risk California carries the largest funded exposure and charged-off amount, while Florida has a higher bad-loan rate. A state dashboard can separate concentration risk from default-rate risk.
06	Use verification status as a diagnostic field Observed bad rates differ by verification status. The next step should be to test whether borrower mix, grade, term or purpose explains the difference.
07	Monitor growth quality, not only growth Applications and funded amount rise strongly through the year. As volume grows, track bad-loan rate, charged-off amount and repayment together so growth does not hide deterioration in portfolio quality.

8. Power BI Dashboard Walkthrough

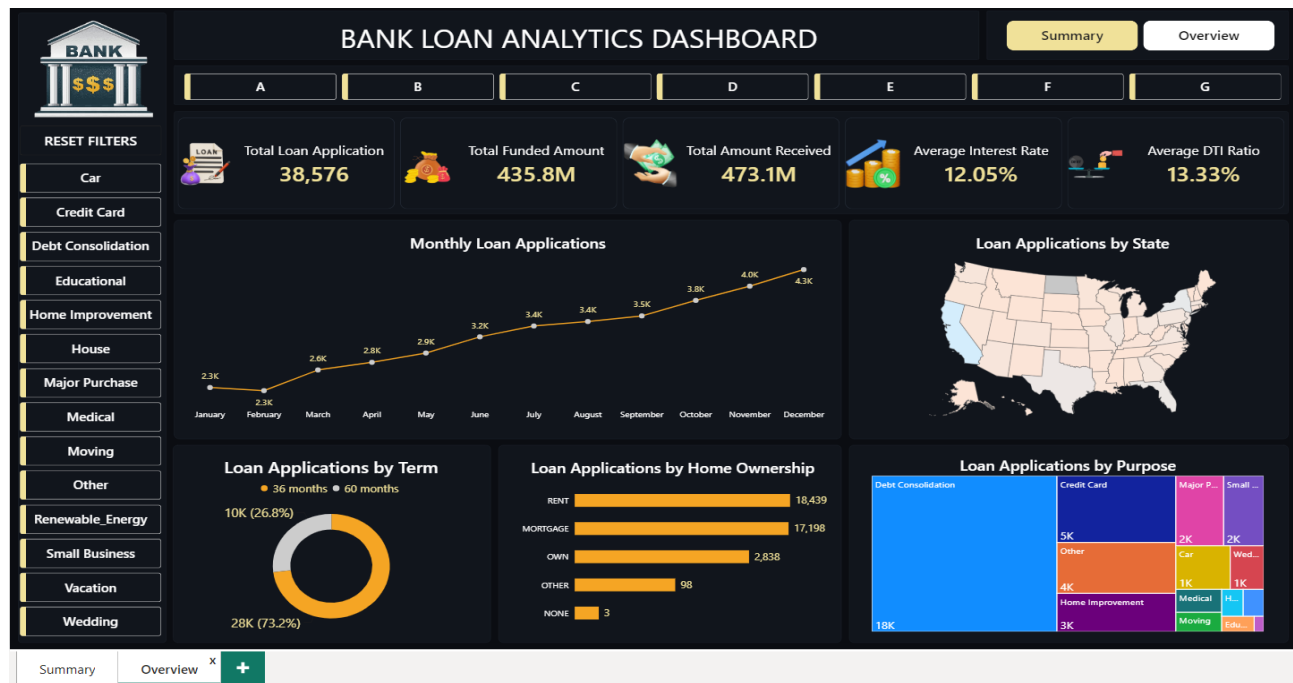
The dashboard is designed as a two-page report: Summary for portfolio health and Overview for deeper segmentation.

Summary page



The Summary page focuses on Good vs Bad loans, funded amount versus repayment, loan status, average interest rate and average DTI. This matches the supplied project requirement for donut charts, status comparisons and KPI monitoring.

Overview page



The Overview page adds the monthly application trend, state map, loan-term split, home-ownership analysis and purpose treemap. The supplied requirements specifically call for these visuals plus purpose/grade filters, navigation buttons and a reset-filter control.

9. How the Analysis Connects

The main strength of the project is that the dashboard is backed by actual analytical questions.

Step	My work	Output
1. Understand the data	Checked structure, record count, duplicates, missing values and date range.	Cleaned data
2. Explore portfolio	Used SQL to inspect status, purpose, grade, term, owners, purpose, term and state.	Portfolio profile
3. Quantify performance	Calculated total loan amount, total payments, averages and KPIs.	Core KPIs
4. Study risk	Compared bad-loan rates by grade, purpose, term and volume.	Risk segments
5. Build dashboard	Converted the main findings into Power BI KPIs and visualizations.	Interactive report
6. Interpret	Connected high-volume segments with high-risk segments.	Business recommendations

Examples of SQL logic used

- Monthly portfolio movement using DATE_TRUNC on ISSUE_DATE.
- Bad-loan rate using CASE WHEN LOAN_STATUS = 'Charged Off' THEN 1 ELSE 0 END divided by total loans.
- Grade-level interest rate and DTI comparisons using AVG and GROUP BY.
- State-level exposure using COUNT and SUM(LOAN_AMOUNT).
- Purpose and term risk segmentation using charged-off counts and rates.

This approach keeps the dashboard tied to questions that a lending or portfolio team could actually ask, instead of using charts only because they look good.

10. Final Takeaways

What I would carry forward from this project.

- The portfolio contains 38,576 applications with about 435.8M funded and 473.1M received.
- Non-charged-off loans represent 86.18% of applications in the dashboard's Good definition; charged-off loans represent 13.82%.
- Credit grade shows a strong observed relationship with charged-off rate, increasing from Grade A to Grade G.
- 60-month loans have roughly double the charged-off rate of 36-month loans in this dataset.
- Debt consolidation is the largest purpose by volume and charged-off amount, while small business has the highest purpose-level bad rate.
- California has the largest funded exposure and charged-off amount; Florida shows a higher observed bad rate.
- Application volume grows strongly through the year, from 2,332 in January to 4,314 in December.

Possible next version

If I extend this project, I would add a borrower-risk segmentation layer, cohort analysis by issue month, more detailed repayment behavior, and an alert-style page for high-risk combinations such as grade + term + purpose + DTI. I would also add a dedicated data dictionary and a short SQL-to-Power-BI documentation section so another analyst can reproduce the work.

Portfolio positioning: This project demonstrates SQL querying, data quality checks, KPI development, risk segmentation, trend analysis and Power BI storytelling in one end-to-end workflow.

Prepared by Sujal Mondal | Data Analyst Portfolio